

ELECTROPLATING OF COPPER METAL ON IRON VESSELS FOR DOMESTIC APPLICATION.

Exp. No. :

Date :

AIM/ OBJECTIVE:

To coat the copper on Stainless Steel object (SS object) using electrolysis method and estimate amount of copper deposited on the SS objects with different voltage.

PRINCIPLE:

Redox reactions, or oxidation-reduction reactions, describe the transfer of electrons between two chemical species. These reactions are made up of two half-reactions which always occur simultaneously. The reduction half reaction represents the gain of electrons to the oxidizing agent. The oxidation half reaction represents the loss of electrons to the reducing agent.

Faraday's Law: Using Faraday's Law, the amount of anode material deposited on the cathode can be calculated:

$$\text{Mass deposited} = \frac{I \times t \times M}{F \times n}$$

I represents the amount of current (in Amperes), t is the deposition time (in seconds), M is the molar mass of the anode material, F is Faraday's constant ($96,500 \text{ A}\cdot\text{s/mol}$) and n is the number of moles of electrons per mole of anode material (i.e. the oxidation number).

MATERIALS REQUIRED:

0.1M Copper sulphate, 0.5M HCl, Power source, Alligator clips, Electrical Tape, Beakers, Copper rod and SS object.

PROCEDURE:

- Clean your copper rod and SS object with steel wool.
- Weigh copper rod and SS object, record the masses.
- Connect the copper rod to the positive terminal and SS object to negative terminal.
- Obtain 50 ml of 0.1M copper sulfate solution, add ~ 7 ml of 0.5M HCl to solution.
- Place the above set-up in 100ml beaker having copper sulphate and HCl solution.
- Set voltage to achieve a current of ~0.4 amps.
- Remove coin and nail after 5 minutes, rinse with DI bottle, rinse with acetone, and carefully pat dry with paper towel.
- Record mass of cathode and anode.
- The same will be repeated with different voltage of power source.

OBSERVATIONS/ INFERENCE:

S.No.	Voltage of the battery (V)	Before electroplating		After electroplating		Weight of copper coated on SS object (g)
		Weight of copper rod (g)	Weight of SS object (g)	Weight of copper rod (g)	Weight of SS object (g)	

RESULTS & DISCUSSION:

The amount of copper coated in SS objects are: _____, _____ & _____ g

MAPPING OF PO & PSO: (For all the COs covered by this experiment)

CO No	PO1	PO2
1	2	1

ASSESSMENT:

Particulars	Max. Marks	Marks awarded
Preparation	10	
Conduct of Experiment	30	
Results & Discussion	30	
Viva - Voce	20	
Report	10	
Total	100	
Evaluator's Signature		